

HDOC Day-19

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Batch- 15

Implementation of SGPA and CGPA Calculation Using Nested Loops & Decision-Making Constructs in C

Below is a report-ready solution following the given algorithm, including the **test-case table**, **C program**, **compilation**, **execution**, and **testing**.

1. Problem Understanding

The program calculates:

- **SGPA (Semester Grade Point Average)** for each semester.
- **CGPA (Cumulative Grade Point Average)** after all semesters.
- Each semester can contain a different number of courses.
- Invalid inputs must be rejected and entered again.
- Nested loops are used because there are **multiple semesters**, and each semester contains **multiple courses**.
- Decision-making constructs are used for validation and grade-to-grade-point conversion.

Grade Point Conversion

Grade	Grade Point (GP)
O	10
A	9
B	8
C	7
D	6
E	5
P	4

2. Test Case Table

The following cases cover **valid, invalid, and boundary inputs**.

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1 – Valid, 1 semester	ID=1001, m=1, n=2; Course 101: credit=4, grade=O;	SGPA = 9.57, Total Credits = 7, CGPA =	SGPA = 9.57, Total Credits = 7, CGPA =	Pass
2 – Valid, multiple	ID=590032842, m=2; Sem 1: (101,4,O), (102,3,A),	Sem 1 SGPA=9.22, Sem 2 SGPA=7.50,	Sem 1 SGPA=9.22, Sem 2 SGPA=7.50,	Pass
3 – Invalid Global ID	ID=0, then 1001	Error for 0 and accept 1001	Error displayed and valid ID accepted	Pass
4 – Invalid semesters	ID=1001, m=-2, then m=1	Error for -2 and accept 1	Error displayed and 1 accepted	Pass
5 – Invalid number of	n=0, then n=1	Error for 0 and accept 1	Error displayed and 1 accepted	Pass
6 – Invalid course code	code=-101, then 101	Error for -101 and accept 101	Error displayed and 101 accepted	Pass
7 – Invalid credit	credit=0, then 4	Error for 0 and accept 4	Error displayed and 4	Pass
8 – Invalid grade	grade=X, then A	Error for X and accept A	Error displayed and A	Pass
9 – Upper boundary	ID=1001, m=1, n=1, code=101, credit=1, grade=O	SGPA=10.00, CGPA=10.00	SGPA=10.00, CGPA=10.00	Pass
10 – Lower passing	ID=1001, m=1, n=1, code=101, credit=1, grade=P	SGPA=4.00, CGPA=4.00	SGPA=4.00, CGPA=4.00	Pass

3. C Program

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    int global_id, m, n;
    int semester, course;
    int code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC, cumulative_TGC;
    char grade;
    float SGPA, CGPA;

    /* Input and validation of Global ID */
    do
    {
```

```

    printf("Enter Global ID: ");
    scanf("%d", &global_id);

    if (global_id <= 0)
    {
        printf("Invalid Global ID. Enter a positive
value.\n");
    }

    } while (global_id <= 0);

/* Input and validation of number of semesters */
do
{
    printf("Enter number of semesters: ");
    scanf("%d", &m);

    if (m <= 0)
    {
        printf("Invalid number of semesters. Enter a
positive value.\n");
    }

    } while (m <= 0);

/* Initialize cumulative totals */
cumulative_TC = 0;
cumulative_TGC = 0;

/* Loop for semesters */
for (semester = 1; semester <= m; semester++)
{
    semester_TC = 0;
    semester_TGC = 0;

    printf("\n--- Semester %d ---\n", semester);

/* Input and validation of number of courses */
do
{
    printf("Enter number of courses in semester %d:
",
        semester);
    scanf("%d", &n);

```

```

        if (n <= 0)
        {
            printf("Invalid number of courses. "
                   "Enter a positive value.\n");
        }

    } while (n <= 0);

    /* Loop for courses */
    for (course = 1; course <= n; course++)
    {
        printf("\nCourse %d\n", course);

        /* Input and validation of course code */
        do
        {
            printf("Enter course code: ");
            scanf("%d", &code);

            if (code <= 0)
            {
                printf("Invalid course code. "
                       "Enter a positive value.\n");
            }

        } while (code <= 0);

        /* Input and validation of credit */
        do
        {
            printf("Enter course credit: ");
            scanf("%d", &credit);

            if (credit <= 0)
            {
                printf("Invalid credit. "
                       "Enter a positive value.\n");
            }

        } while (credit <= 0);

        /* Input and validation of grade */
        do
        {
            printf("Enter grade (O/A/B/C/D/E/P): ");

```

```

scanf(" %c", &grade);

grade = toupper(grade);

if (grade != 'O' &&
    grade != 'A' &&
    grade != 'B' &&
    grade != 'C' &&
    grade != 'D' &&
    grade != 'E' &&
    grade != 'P')
{
    printf("Invalid grade. Enter O, A, B, "
           "C, D, E, or P.\n");
}

} while (grade != 'O' &&
        grade != 'A' &&
        grade != 'B' &&
        grade != 'C' &&
        grade != 'D' &&
        grade != 'E' &&
        grade != 'P');

/* Find Grade Point */
switch (grade)
{
    case 'O':
        GP = 10;
        break;

    case 'A':
        GP = 9;
        break;

    case 'B':
        GP = 8;
        break;

    case 'C':
        GP = 7;
        break;

    case 'D':
        GP = 6;

```


4. Sample Execution

```
laaryanrajput@aaryans-macbook ~ % nano day19.c
laaryanrajput@aaryans-macbook ~ % clang day19.c -o day19.c
laaryanrajput@aaryans-macbook ~ % ./day19.c
Enter Global ID: 590832842
Enter number of semesters: 3

--- Semester 1 ---
Enter number of courses in semester 1: 3

Course 1
Enter course code: 101
Enter course credit: 102
Enter grade (O/A/B/C/D/E/P): o

Course 2
Enter course code: 102
Enter course credit: 100
Enter grade (O/A/B/C/D/E/P): P

Course 3
Enter course code: 103
Enter course credit: 98
Enter grade (O/A/B/C/D/E/P): p

Semester 1 SGPA = 6.84

--- Semester 2 ---
Enter number of courses in semester 2: 2

Course 1
Enter course code: 104
Enter course credit: 100
Enter grade (O/A/B/C/D/E/P): o

Course 2
Enter course code: 105
Enter course credit: 100
Enter grade (O/A/B/C/D/E/P): a

Semester 2 SGPA = 9.50

--- Semester 3 ---
Enter number of courses in semester 3: 2

Course 1
Enter course code: 107
Enter course credit: 100
Enter grade (O/A/B/C/D/E/P): a

Course 2
Enter course code: 108
Enter course credit: 100
Enter grade (O/A/B/C/D/E/P): b

Semester 3 SGPA = 8.50

===== FINAL RESULT =====
Global ID      = 590832842
Number of Semesters = 3
Cumulative Total Credits = 700
CGPA           = 7.73
laaryanrajput@aaryans-macbook ~ %
```

The program was **compiled and executed successfully**, and the calculated SGPA/CGPA matched the expected results for the tested use case.